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Final set of equations :-

1 cabin bounce (z_c) :-

$$m_c \ddot{z}_c + (c_{cfL} + c_{cfr} + c_{crl} + c_{crr})\dot{z}_c$$
$$- (c_{cfl} + c_{cfr} + c_{crl} + c_{crr})\dot{z}_s$$
$$- (-k_{cfl} - k_{cfr} - k_{crl} - k_{crr})z_c$$
$$- (k_{cfl} + k_{cfr} + k_{crl} + k_{crr})z_s$$
$$- (c_{cfl} \times l_{cfcg} + c_{cfr} \times l_{cfcg} - c_{crl} \times l_{crcg}$$
$$- c_{crr} \cdot l_{crcg})\dot{\theta}_c$$
$$- (-c_{cfl} \times l_{cf} - c_{cfr} l_{cf} - c_{crl} \cdot l_{cr} - c_{crr} l_{cr})\dot{\theta}_s$$
$$- (-c_{cfl} \times b + c_{cfr} \times a - c_{crl} b + c_{crr} \times a)\dot{\phi}_c$$
$$- c_{cfl} \times b - c_{fr} a + c_{crl} b - c_{crr} \times a)\dot{\phi}_s$$

$$- (k_{cfl} \times l_{cfcg} + k_{cfr} \times l_{cfcg} - k_{crl} l_{crcg}$$
$$- k_{crr} \times l_{crcg})\theta_c$$
$$- (-k_{cfl} \times l_{cf} - k_{cfr} \times l_{cf} - k_{crl} \times l_{cr} - k_{crr} \times l_{cr})$$
$$\times \theta_s$$
$$- (-k_{cfl} \times b + k_{cfr} \times a - k_{crl} \times b$$
$$+ k_{crr} \times a)\phi_c$$
$$- (k_{cfl} \times b - k_{cfr} \times a + k_{crl} \times b - k_{crr} \times a)$$
$$\phi_s$$
$$= 0$$

$$= (z_{us})_{2L}$$

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2. Cabin pitch (θ_c)

$$I_{yyc}\ddot{\theta}_c - (c_{cfl} * l_{cfcg} + c_{cfr} l_{cfcg} - c_{crl} * l_{crcg} - c_{crr} * l_{crcg})\dot{z}_c$$
$$- (-c_{cfl} * l_{cfcg} - c_{cfr} * l_{cfcg} - c_{crl} * l_{crcg} - c_{crr} * l_{crcg}) * \dot{z}_s$$
$$- (-c_{cfl} * l_{cfca}^2 - c_{cfr} * l_{cfcg}^2 - c_{crl} * l_{crcg}^2 - c_{crr} l_{crcg}^2)\dot{\theta}_c$$
$$- (c_{cfl} * l_{cfcg} * l_{cf} + c_{cfr} * l_{cfcg} * l_{cf} - c_{crl} * l_{crcg} * l_{cr} - c_{crr} * l_{crcg} * l_{cf})\dot{\theta}_s$$
$$- (-c_{cfl} * l_{cfcg} * b + c_{cfr} * l_{cfcg} * a - c_{crl} * l_{crcg} * b + c_{crr} * l_{crcg} * a)\dot{\phi}_c$$
$$- (c_{cfl} * l_{cfcg} * b - c_{cfr} * l_{cfcg} * a + c_{crl} * l_{crcg} * b - c_{crr} * l_{crcg} * a)\dot{\phi}_s$$
$$- (k_{cfl} * l_{cfcg} + k_{cfr} * l_{cfcg} - k_{crl} * l_{crcg} - k_{crr} * l_{crcg}) * z_s$$
$$- (-k_{cfl} * l_{cfcg} - k_{cfr} * l_{cfcg} - k_{crl} * l_{crcg} - k_{crl} * l_{crcg}) z_s$$
$$- (-k_{cf} * l_{cfcg}^2 - k_{cfr} * l_{cfcg}^2 - k_{crl} * l_{crcg}^2 - k_{crr} * l_{crcg}^2 + m_c * g * h_{cp})\theta_c$$
$$- (k_{cfl} * l_{cfcg} * l_{cf} + k_{cfr} * l_{cfcg} * l_{cf} - k_{crl} * l_{crcg} * l_{cr} - k_{crr} * l_{crcg} * l_{cf})\theta_s$$
$$- (-k_{cfl} * l_{cfcg} * b + k_{cfr} * l_{cfcg} * a - k_{crl} * l_{crcg} * b + k_{crr} * l_{crcg} * a)\phi_c$$
$$- (k_{cfl} * l_{cfcg} * b - k_{cfr} * l_{cfcg} * a + k_{crl} * l_{crcg} * b - k_{crr} * l_{crcg} * a)\phi_s = 0$$

Cabin Roll (ϕ_c) :-

$$I_{xx_c}\ddot{\phi}_c - (-c_{fl} * b + c_{fr} * a - c_{rl} * b + c_{rr} * a)\dot{z}_c$$
$$- (c_{fl} * b - c_{fr} * a + c_{rl} * b - c_{rr} * a)\dot{z}_s$$
$$- (-c_{fl} * l_{fcg} * b - c_{fr} * l_{fcg} * a + c_{rl} * l_{rcg} * b + c_{rr} * l_{rcg} * a)\dot{\theta}_c$$
$$- (c_{fl} * l_{fcg} * b + c_{fr} * l_{fcg} * a - c_{rl} * l_{rcg} * b - c_{rr} * l_{rcg} * a)\dot{\theta}_s$$
$$- (-c_{fl} * b^2 + c_{fr} * a^2 - c_{rl} * b^2 + c_{rr} * a^2)\dot{\phi}_c$$
$$- (c_{fl} * b^2 - c_{fr} * a^2 + c_{rl} * b^2 - c_{rr} * a^2)\dot{\phi}_s$$
$$- (-k_{fl} * b + k_{fr} * a - k_{rl} * b + k_{rr} * a) z_c$$
$$- (k_{fl} * b - k_{fr} * a + k_{rl} * b - k_{rr} a) z_s$$
$$- (-k_{fl} * l_{fcg} * b - k_{fr} * l_{fcg} * a + k_{rl} * l_{rcg} * b + k_{rr} * l_{rcg} * a)\theta_c$$
$$- (k_{fl} * l_{fcg} * b + k_{fr} * l_{fcg} * a - k_{rl} * l_{rcg} * b - k_{rr} * l_{rcg} * a)\theta_s$$
$$- (-k_{fl} * b^2 + k_{fr} * a^2 - k_{rl} * b^2 + k_{rr} * a^2)\phi_c$$
$$- (k_{fl} * b^2 - k_{fr} * a^2 + k_{rl} * b^2 - k_{rr} * a^2)\phi_s = 0$$

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Sprung mass bounce :- $(a \Rightarrow l_1)$

$$M_s \ddot{z}_s + K_f (z_s - a\theta_s - z_{uf}) + C_f (\dot{z}_c - a\dot{\theta} - \dot{z}_{uf})$$

$$+ k_2 (z_s - z_2 - \beta_{L2} L_{DL2} - \beta_{R2} L_{DR2} + l_2\theta)$$
$$+ C_2 (\dot{z}_s - \dot{z}_2 + l_2\dot{\theta})$$

$$+ k_3 (z_s - z_3 - \beta_{L3} L_{DL3} - \beta_{R3} L_{DR3} + l_3\theta)$$
$$+ C_3 (\dot{z}_s - \dot{z}_3 + l_3\dot{\theta}) = 0$$

Sprung mass Pitch :-

$$I_{yy}\ddot{\theta} + I_{yx}\ddot{\phi}_s - a[k_f (z_s - a\theta - z_{uf}) + C_f (\dot{z}_c - a\dot{\theta} - \dot{z}_{uf})]$$

$$+ l_2 [k_2 (z_s - z_2 - \beta_{L2} L_{DL2} - \beta_{R2} L_{DL2} + l_2\theta)$$
$$+ C_2 (\dot{z}_s - \dot{z}_2 + l_2\dot{\theta})]$$

$$+ l_3 [k_3 (z_s - z_3 - \beta_{L3} L_{DL3} - \beta_{R3} L_{DL3} + l_3\theta)$$
$$+ C_3 (\dot{z}_s - \dot{z}_3 + l_3\dot{\theta})]$$

Sprung mass Roll :-

$$-(I_{xx}\ddot{\phi}_s + m_2 h_1^2 \ddot{\phi}_s) + I_{xy}\ddot{\theta} = -m_2 g h_1 \phi_s$$

$$+ k_{rf} (\phi_s - \phi_f) + C_{rf} (\dot{\phi}_s - \dot{\phi}_f)$$

$$+ k_{r1} (\phi_s - \phi_{r1} - \phi_{NRS}) + C_{r1} (\dot{\phi}_s - \dot{\phi}_{r1})$$

$$+ k_{r2} (\phi_s - \phi_{r2} - \phi_{NRS}) + C_{r2} (\dot{\phi}_s - \dot{\phi}_{r2})$$

$i = 2, 3$ $\quad \phi_{NRS} = \dfrac{\beta_{Li} L_{DLi} - \beta_{Ri} L_{DRi}}{s_{tfi}}$

Front axle - bounce :-

$$M_{1f}\ddot{z}_{1f} + k_f(z_{1f} - (z_s - a\theta))$$
$$+ c_f(\dot{z}_{1f} - (\dot{z}_s - a\dot{\theta})) = 0.$$

Front axle roll :-

$$-(I_{xx}\ddot{\phi}_f + m_f h_2^2 \ddot{\phi}_f)$$
$$= -m_f g h_{us} (\phi i) + k_{rf}(\phi_f - \phi_s)$$
$$+ c_{rf}(\dot{\phi}_f - \dot{\phi}_s)$$

Rear - r_1 bounce :-

$$M_2\ddot{z}_2 - k_2(z_s - z_2 - \beta_{L2} L_{DL2} - \beta_{R2} L_{DR2} + l_2\theta)$$
$$- c_2(\dot{z}_s - \dot{z}_2 + l_2\dot{\theta}) = 0$$

Rear - r_2 roll :-

$$-(I_{xx}\ddot{\phi}_2 + m_2 h_2^2 \ddot{\phi}_2)$$
$$= -m_2 g h_{us}(\phi_2 + \phi_{NRS}) + k_{r2}(\phi_2 + \phi_{NRS} - \phi_s)$$
$$+ c_{r2}(\dot{\phi}_2 - \dot{\phi}_s)$$

Rear axle - r_2 bounce :-

$$M_3\ddot{z}_3 - k_3(z_s - z_3 - \beta_{L3} L_{DL3}$$
$$- \beta_{R3} L_{DR3} + l_3\theta)$$
$$- c_2(\dot{z}_s - \dot{z}_3 + l_3\dot{\theta}) = 0$$

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Rear axle - roll - r_2 :-

$$-(I_{xx}\ddot{\phi}_3 + m_3 h_2^2 \ddot{\phi}_3)$$

$$= -m_3 g h_{us}(\phi_3 + \phi_{NRS}) + k_{r3}(\phi_3 + \phi_{NRS} - \phi_s) + c_{r3}(\dot{\phi}_3 - \dot{\phi}_s)$$

$$(z_s + l_{i2}\theta + S_{tf_2}\phi) - (L_D + \Delta l)\sin(\beta_{2L} - \theta) = (z_{us})_{2L}$$

$$(z_s + l_{i3}\theta + S_{tf_2}\phi) - (L_D + \Delta l)\sin(\beta_{3L} - \theta) = (z_{us})_{3L}$$